2027 MATHEMATICAL CONTEST IN MODELING — PRACTICE

Problem A: Where Does the Sand Go?

Can Beach Nourishment Keep Up with a Changing Coastline?

Background

The Outer Banks of North Carolina are a chain of barrier islands separating the Atlantic Ocean from coastal sounds and communities. Their sandy shorelines are continually reshaped by waves, currents, storms, and sediment transport. A beach can become narrower during a storm, regain some sand afterward, or gradually change position over many years. These processes occur on different time scales and may redistribute sand along the shore rather than simply remove it from the coastal system.

The consequences are tangible near **Rodanthe, North Carolina**, where coastal erosion has placed oceanfront structures at risk. Dare County documents houses that have collapsed along the oceanfront; these events demonstrate the seriousness of local exposure but do not, by themselves, establish a universal erosion rate or a complete history of the shoreline. North Carolina publishes long-term shoreline-change rates used in coastal management, while the U.S. Geological Survey (USGS) provides broader coastal-change information. Such historical rates are not necessarily reliable forecasts for a particular storm or a particular decade.

One response is *beach nourishment*: placing additional sand on a beach to widen it or replenish its sediment supply. Yet the ocean continues to move that sand. A project that appears successful immediately after placement may perform differently after several months of waves, one intense storm, or a sequence of storms. Nourishment at one location may also influence the sediment available farther along the coast. Repeating the project too often may be expensive, while waiting too long may expose valuable infrastructure.

Suppose a **hypothetical coastal resilience task force** is considering sand-placement strategies for a **representative 5-km coastal corridor near Rodanthe** during **2027–2042**. The corridor, its endpoints, and any intervention budget used in this problem are *contest assumptions*, not an approved engineering project or a defined local government planning area. The task force asks your team:

When does adding sand truly protect a changing shoreline, and when does the ocean simply move the problem elsewhere?

Your Challenge

Develop a **continuous, time-dependent model** of beach evolution and a defensible coastal-management recommendation. Use the Outer Banks as your primary case study. You may use observed data where available and construct transparent illustrative scenarios when the measurements required by your model are not available. Make the physical meaning, dimensions, and limitations of all state variables and parameters clear.

1. Describe the beach before intervention. Choose measurable indicators of coastal condition, such as shoreline position, dry-beach width, beach-profile shape, or sediment volume. Develop a model for how these quantities change continuously in space and time under ordinary waves and alongshore sediment transport, while allowing for temporary changes caused by storms and subsequent recovery. Justify your model boundary conditions and explain how the chosen indicators relate to risk for nearby infrastructure. Distinguish long-term shoreline trends from short-term changes during individual storms.

2. Follow the sand. Represent the addition of sand to part of the corridor and predict how its distribution evolves afterward. Compare at least two placement patterns with the same total initial sand volume. Account for transport into neighboring reaches, possible offshore losses, and any recovery processes you consider important. Explain where the added sand goes in your model and demonstrate whether your sediment accounting is internally consistent. Identify conditions under which a wide initial beach does *not* produce lasting protection.

3. Connect the model to evidence. Estimate or bound important parameters using credible public data, published physical relationships, or clearly labeled assumptions. Test the model against historical shoreline observations or other evidence not used for calibration, where feasible. Assess sensitivity, parameter identifiability, and the effects of imperfect data. If the available observations cannot distinguish between competing physical mechanisms, explain what additional measurements would be needed.

4. Challenge the shoreline with future conditions. Evaluate the evolution of the corridor during **2027–2042** under at least three contrasting conditions: a relatively calm sequence of years, a major storm soon after nourishment, and repeated storms with changing relative sea level or wave climate. Quantify both the duration of useful protection and uncertainty in the predicted shoreline or beach width. Explain how your conclusions change if the timing of storms is different, even when their total number is similar.

5. Recommend a strategy that can adapt. Compare a single large nourishment project, smaller repeated placements, and spatially targeted nourishment under a defensible budget or sand-supply constraint. Develop a rule for *when* and *where* to intervene based on observable coastal conditions rather than a fixed calendar alone. Consider when nourishment may be inappropriate and another response, such as relocating vulnerable assets, deserves consideration. Recommend monitoring indicators that would cause the task force to revise the strategy.

What to Submit

Submit an English mathematical modeling report of **no more than 25 pages** for this practice contest. Include a one-page summary, physical assumptions, governing relationships, parameter choices and units, numerical or analytical results, validation or plausibility checks, uncertainty analysis, and an actionable recommendation. Include a **one-page technical letter to the hypothetical coastal resilience task force** explaining what to do first, how to monitor success, and when to reconsider the plan.

Important distinctions: A long-term shoreline-change rate is not a storm-by-storm forecast; shoreline position, beach width, and sediment volume are not interchangeable; storm-driven erosion may be followed by recovery; regional sea-level measurements are not direct local shoreline-retreat measurements; and a numerical scenario is not evidence that an unmeasured intervention has occurred. Keep observed data, modeled projections, and hypothetical interventions separate.